

B

PART B Lighting Design

The R13 colour-forward design on the venue's installed 39-fixture rig: rig, layers, specials, the look library and song colour as programmed, fixture schedule and focus plan.

Revision R13.1 · 28 September 2026

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Lighting Design

R13 FLASHY SCENE SPLIT — readable faces in neutral FOH light, with the energy coming from changing LX1 colour, side-light contrast, LX2 backlight and animated PixBars.

39 lighting fixtures + hazer, 4 types	153 programmed cue positions	17 looks M01–M17 (P6–P7)	10 song memories (P4)	0 strobe values stored
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WHAT CHANGED FROM R8

R8 lit the show from one 87-step list on Page 2 Memory 1 with mostly naturalistic looks. R13 keeps the rig, patch and focus but rebuilds the programming: **stronger LX1 colour, side light, LX2 backlight and PixBar movement**, fast colour snaps on musical impacts, slower crossfades for emotional and dialogue moments, and every song has its own colour-driven memory. The show is split across pages: **P2 Act One, P3 Act Two, P4 songs, P5 FX/chases, P6–P7 look library, P8 full 153-step backup**. QLab fires every cue; R13.1 adds automatic flash returns and memory releases (Part D).

1 The installed rig

Fixture	Qty	Mantra #	DMX	What it is	Job in R13
Lightsky 8800-C42	12	#1–12	U2 · 11 ch · 101–232	LED profile with shutters, 6 colours + CCT, 19° lens	Face light and hard-edged specials. Best skin tones.
Tour Pro Zoom	10	#13–22	U1 · 12 ch · 301–420	RGBW COB zoom wash 13–45°, barn doors	Colour top wash (LX1), voice-transfer pools, high side
Tour Pro TourCOB PAR	11	#23–32, 39	U1 · 6 ch · 421–486	RGBW COB PAR, fixed beam	Backlight and side light; the main colour energy of R13
Tour Pro PixBar	6	#33–38	U1 · 6 ch · 1–36	Linear bar, whole-bar colour in this mode	Effect row: storm, magic, jellyfish, song impacts
Hazer	1	#40	U1 · 2 ch · 487–488	Output + fan	Selective haze, P5 M4 (50 % / fan 50 %)

Positions are unchanged from R8: 8 × C42 on the FOH truss (#1–8), LX1 downstage bar (C42 #9–12, Zoom #13–18), LX2 upstage bar (Zoom #19–20, COB #23–28, 39, PixBar #33–38), a floor boom in each wing (SR #21, 29, 30 · SL #22, 31, 32) and the hazer upstage on the floor (#40). The projector hangs centre stage towards the back and throws upstage (R-11).

DESIGN RULE

FOH #1–6 keep faces readable in every look. The flashier colour movement comes from the LX1 Zooms, side booms, LX2 backlight and PixBars. Never aim high-intensity backlight or PixBars at the projection screen (R-25).

2 Design layers

Layer	How it is set in R13	What it does
Face (FOH C42 #1–6)	Near-white: cool white under water, warm white on land (about 30–100 %)	Every scene with dialogue stays readable
Colour top (LX1 Zoom #13–18)	Saturated scene colour: cyan, hot pink, gold, blue, violet	Sets the scene colour; snaps on musical impacts
Side (booms #21–22, 29–32)	Contrasting colour to the top wash (lime, violet, magenta, teal)	Body light and colour contrast for dance, off the screen
Backlight (LX2 COB #23–28, 39)	Highest levels in the rig (60–100 %), strong saturated colour	Depth, haze beams, the storm and lair energy
Effects (PixBar #33–38)	Two-colour splits in storm, magic, jellyfish and every song	Movement and impact without strobe
Specials (C42 #7–12, Zoom #19–20)	Hard pools on marks, added per cue	Character entrances, shell, voice transfer

Colour language by scene family

Scene family	R13 colour language	Looks · cues
Underwater	Deep blue → cyan → turquoise → purple → blue/green	M03 · Q4–Q11, Q31
Comedy	Aqua → hot pink → yellow/gold → purple accents	M04 · Q9, Q12–Q14
Royal / palace	Gold + royal/deep blue, magenta/gold musical lifts	M05 · Q10, Q39–Q44
Ship / storm	Steel/deep blue + amber; storm through cyan/purple/ice-white impacts	M06, M08, M09 · Q18–Q27
Shore / dry land	Amber → peach → pink → lavender → moonlight blue	M07, M14 · Q28–Q30.5, Q59
Magic / Spirit	Purple → cyan → teal/green → white impact	M10 · Q33–Q36, Q48, Q62
Lair / Octavia	Purple + acid green → red/magenta → cyan/purple	M11 · Q50–Q58.5
Jellyfish	Hot pink + cyan + violet pulse	M13 · Q45–Q49
Wedding / finale	Gold → aqua → magenta → blue → gold / multicolour	M15, M16 · Q60–Q65

NO STROBE

No strobe is programmed: Strobe, Control, Reset, Auto speed, Other, Default and Colour macro are 0 in every cue. Lightning and magic are single white hits (Q21, Q23, Q25, Q35, Q57b) that QLab returns automatically after 0.2–0.3 s. Never more than 3 flashes a second (R-04); the HAYWIRE chase runs at 2 steps a second.

3 Specials

Special	Character	Fixture	Use
SP1	Ariel	#7 C42 (FOH)	DSC mark; solo, duet, transformation, restore (57c)
SP2	Spirit	#8 C42 (FOH)	DS audience-left mark; every Spirit entrance
SP3	Octavia	#9 C42 (LX1)	US audience-right; entrances, lair, wedding interruption
SP4	Shell	#10 C42 (LX1, shuttered tight)	Plinth mark; brightest point in Scene Eight
V1	Dame	#11 C42 (LX1)	Voice-transfer mark 1 (Q53)
V2	Flanders	#12 C42 (LX1)	Voice-transfer mark 2 (Q54)
V3	Theodore	#19 Zoom 13° + barn doors (LX2)	Voice-transfer mark 3 (Q55)
V4	Marina	#20 Zoom 13° + barn doors (LX2)	Voice-transfer mark 4 (Q56)

4 Look library M01–M17 — as programmed

The looks live on **Page 6 (M01–M10)** and **Page 7 (M11–M17)** for rebuilding and rehearsal. Performances run the scene memories on P2/P3 and the song memories on P4, which were built from these looks. Values below are read from the R13.1 show file.

Swatch order: **Face** (FOH #1–6) · **Top** (LX1 #13–18) · **Side** (booms) · **Back** (LX2 COB) · **Pix** (PixBars) · **Spec** (specials). Colour as programmed in the R13.1 show file, darker = lower level, black = off.

Look	Desk	Layers	As programmed (level · colour)
M01 BLACKOUT	P6 M1	     	all out
M02 HOUSE	P6 M2	     	Back 10% blue
M03 UNDERWATER	P6 M3	     	Face 55% cool white · Top 55% cyan · Side 60% lime · Back 88% red
M04 COMEDY	P6 M4	     	Face 85% warm white · Top 72% hot pink · Side 68% violet · Back 62% sky blue
M05 ROYAL	P6 M5	     	Face 75% warm white · Top 65% gold · Side 60% hot pink · Back 82% violet
M06 SHIP	P6 M6	     	Face 70% warm white · Top 58% blue · Side 55% teal · Back 80% orange
M07 SHORE	P6 M7	     	Face 75% orange · Top 62% cyan · Side 58% lime · Back 66% red
M08 STORM	P6 M8	      	Face 35% cool white · Top 28% violet · Side 48% blue · Back 100% teal · Pix 72% warm white + lime
M09 LIGHTNING	P6 M9	      	Face 100% cool white · Top 100% sky blue · Side 100% cyan · Back 100% gold · Pix 100% sky blue + red
M10 MAGIC	P6 M10	      	Face 45% deep blue · Top 60% teal · Side 66% orange · Back 92% magenta · Pix 82% lime + violet
M11 LAIR	P7 M1	     	Face 30% cool white · Top 34% violet · Side 46% sky blue · Back 88% green · specials 70%

Look	Desk	Layers	As programmed (level · colour)
M12 VOICE SHELL	P7 M2		Face 5% cool white · Side 16% lime · Back 18% red · specials 85%
M13 JELLYFISH	P7 M3		Face 65% warm white · Top 68% hot pink · Side 70% violet · Back 80% sky blue · Pix 78% magenta + cyan
M14 DRYLAND	P7 M4		Face 80% warm white · Top 60% orange · Side 52% magenta · Back 56% warm white
M15 WEDDING	P7 M5		Face 90% warm white · Top 78% gold · Side 70% hot pink · Back 88% violet
M16 BOWS	P7 M6		Face 100% warm white · Top 88% magenta · Side 82% sky blue · Back 90% cyan
M17 SAFE LIGHT	P7 M7		Face 70% white

CHECK IN THE VENUE

Some R13 backlight choices are deliberately bold (for example the coral-red backlight under M03 UNDERWATER and the teal backlight in M08 STORM). Look at every look from the house with performers at the focus session, and record any change in Manual Appendix B before saving the desk.

5 Songs — Page 4

Each song has its own memory on Page 4. QLab starts the track and the first section together; every later section is one GO on the music (Part E §4).

Swatch order: **Face** (FOH #1–6) · **Top** (LX1 #13–18) · **Side** (booms) · **Back** (LX2 COB) · **Pix** (PixBars) · **Spec** (specials). Colour as programmed in the R13.1 show file, darker = lower level, black = off.

Song	Desk	Colour language	Sections as programmed (one strip per section)
S1 Rock Lobster	P4 M1 · 7 cues	Aqua / hot pink / gold / deep blue / magenta / lime / white impacts	
S2 Feeling Good (Austin Millz Remix)	P4 M2 · 6 cues	Deep blue / purple / gold / magenta with warm-white highlights	
S3 Part of Your World	P4 M3 · 7 cues	Blue / teal / violet / pink / cyan with a warmer emotional lift	
S4 Wellerman (Sea Shanty)	P4 M4 · 7 cues	Steel/deep blue and amber, then teal/cyan and bright chorus accents	
S5 (I've Had) The Time of My Life	P4 M5 · 8 cues	Amber / pink / violet / cyan / gold with a larger final chorus	
S6 Crab Rave	P4 M6 · 7 cues	Fastest colour changes: lime, cyan, magenta, gold, red and white impacts	
S7 Jellyfish chorus (song to be decided)	P4 M7 · 4 cues	Hot pink / cyan / violet pulse language	
S8 Poor Unfortunate Souls	P4 M8 · 7 cues	Purple / acid green / red / cyan for strong villain contrast	
S9 Absolutely Everybody	P4 M9 · 7 cues	Concert style: cyan / magenta / gold / blue / lime / hot pink	
S10 He's a Pirate	P4 M10 · 6 cues	Deep blue / amber / cyan / red / gold / bright white finale	

6 FX and chases — Page 5

Desk	Memory	Programming	Use
P5 M1	WARM CHASE	6 steps @ 150 BPM	Rehearsal / optional party energy
P5 M2	COOL CHASE	6 steps @ 144 BPM	Rehearsal / optional underwater movement
P5 M3	PARTY CHASE	8 steps @ 168 BPM	Optional finale / bows
P5 M4	HAZE	single look	Hazer #40 at 50 %, fan 50 % — run on its fader, 20–40 s before the look
P5 M5	HAYWIRE	5 steps @ 120 BPM	Q57 on, Q57b off — fired by QLab (V1, SP4, V2, V3, V4 at 85 %)
P5 M6	JELLY PULSE	single look	Bump for extra Q47 repeats

7 Fixture schedule and focus plan

One row per fixture. Tick when focused; note the final colour/level. Addresses are the venue patch (Part C).

#	Type	Position	DMX	Focus / role	✓
1	C42	FOH	U2:101	Face zone 1 — DSR (from house left)	☐
2	C42	FOH	U2:112	Face zone 1 — DSR (from house right)	☐
3	C42	FOH	U2:123	Face zone 2 — DSC (from house left)	☐
4	C42	FOH	U2:134	Face zone 2 — DSC (from house right)	☐
5	C42	FOH	U2:145	Face zone 3 — DSL (from house left)	☐
6	C42	FOH	U2:156	Face zone 3 — DSL (from house right)	☐
7	C42	FOH	U2:167	SP1 Ariel — DSC mark, shuttered tight	☐
8	C42	FOH	U2:178	SP2 Spirit — DS audience-left mark	☐
9	C42	LX1	U2:189	SP3 Octavia — US audience-right mark	☐
10	C42	LX1	U2:200	SP4 Shell — plinth mark, shuttered to the shell	☐
11	C42	LX1	U2:211	V1 Dame — voice-transfer mark 1	☐
12	C42	LX1	U2:222	V2 Flanders — voice-transfer mark 2	☐
13	ZM	LX1	U1:301	Top/front colour wash — DSR	☐
14	ZM	LX1	U1:313	Top/front colour wash — DSC	☐
15	ZM	LX1	U1:325	Top/front colour wash — DSL	☐
16	ZM	LX1	U1:337	Top/front colour wash — CSR	☐
17	ZM	LX1	U1:349	Top/front colour wash — CSC	☐
18	ZM	LX1	U1:361	Top/front colour wash — CSL	☐
19	ZM	LX2	U1:373	V3 Theodore — 13°, barn doors, steep top light	☐
20	ZM	LX2	U1:385	V4 Marina — 13°, barn doors, steep top light	☐
21	ZM	SR boom	U1:397	High side — top of boom (~2.8 m), across to SL, head height	☐
22	ZM	SL boom	U1:409	High side — top of boom (~2.8 m), across to SR, head height	☐
23	COB	LX2	U1:421	Backlight — DSR	☐
24	COB	LX2	U1:427	Backlight — DSC	☐
25	COB	LX2	U1:433	Backlight — DSL	☐
26	COB	LX2	U1:439	Backlight — CSR	☐
27	COB	LX2	U1:445	Backlight — CSC	☐
28	COB	LX2	U1:451	Backlight — CSL	☐
29	COB	SR boom	U1:457	Mid side — boom arm ~1.2 m, knee to hip	☐
30	COB	SR boom	U1:463	Low side (shin) — boom arm ~0.4 m	☐
31	COB	SL boom	U1:469	Mid side — boom arm ~1.2 m, knee to hip	☐
32	COB	SL boom	U1:475	Low side (shin) — boom arm ~0.4 m	☐
33	PIX	LX2	U1:1	Effect row, tilted downstage off the screen (R-25)	☐
34	PIX	LX2	U1:7	Effect row, tilted downstage off the screen (R-25)	☐
35	PIX	LX2	U1:13	Effect row, tilted downstage off the screen (R-25)	☐
36	PIX	LX2	U1:19	Effect row, tilted downstage off the screen (R-25)	☐
37	PIX	LX2	U1:25	Effect row, tilted downstage off the screen (R-25)	☐
38	PIX	LX2	U1:31	Effect row, tilted downstage off the screen (R-25)	☐
39	COB	LX2	U1:481	Extra backlight — centre (confirm it exists, R13.1 fix list)	☐
40	HAZE	Floor, US	U1:487	Atmosphere — P5 M4 HAZE (50 %, fan 50 %)	☐
41	PIN	FOH SR end	U1:37	Mirror ball (optional, not patched)	☐
42	PIN	FOH SL end	U1:38	Mirror ball (optional, not patched)	☐

Focus notes

- FOH C42s: cross-light each zone from both sides; shutter off the screen, the proscenium and the front row. Fit the 26° or 36° lenses if the pools don't overlap.
- Specials: shutter each C42 tight to its glow-tape mark. The shell special (#10) is the brightest thing on stage in Scene Eight.
- LX1 Zooms: overlap by a third; barn doors cut the top edge so nothing reaches the screen.
- LX2: COB PARs down-and-forward to head height at the DS edge; PixBars tilted downstage, never onto the screen (R-25). R13 runs the backlight hard, so check screen washout with BG-02 and BG-18 running.
- Side booms: Zoom high at head height across the stage; COB mid and low for bodies and dance.

- Voice transfer: Zooms #19–20 at 13° with barn doors match Dame and Flanders (#11–12); match the four levels by eye.

Designer / focus notes